Sachin Upadhyay

Leetcode | **HackerRank**
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Professional Summary

Data Science and AI enthusiast and B.Tech student at NIT Raipur with a strong foundation in deep learning, natural language processing, and statistical modeling. Proficient in Python, PyTorch, and predictive modeling to architect intelligent systems from large-scale, complex datasets. Experienced in building end-to-end ML pipelines, optimizing neural network performance, and deploying scalable AI solutions. Seeking to leverage data-driven strategies and technical expertise to solve real-world business challenges through cutting-edge Artificial Intelligence and machine learning.

Education

Degree / Board	Institution	Year	CGPA / %
B.Tech, Biotechnology	National Institute of Technology (NIT), Raipur	2023 – 2027	7.9 / 10.0
Senior Secondary (XII) - UPMSP	GSAS Academy, Basti, Uttar Pradesh	2020 – 2021	83.33%
Secondary (X) - UPMSP	GSAS Academy, Basti, Uttar Pradesh	2018 – 2019	83.33%

Skills

Languages & Databases: Python, SQL, MySQL, MongoDB, C++ (Basic)
Data Science & Analysis: Pandas, NumPy, Matplotlib, Seaborn, Streamlit, Exploratory Data Analysis (EDA)
Machine Learning (ML): Scikit-learn, Predictive Modeling, Regression, Classification, Feature Engineering, Cross-Validation
AI, Deep Learning & GenAI: Generative AI (LLMs), Natural Language Processing (NLP), PyTorch, TensorFlow, Neural Networks, Computer Vision
MLOps & Deployment: Docker, AWS (EC2/S3), FastAPI, Git, GitHub, Google Colab, VS Code

Projects

Lung Disease Diagnosis System

November 2025 – January 2026

- **Designed** a deep learning pipeline to automate feature extraction from 5,000+ chest X-ray images, utilizing CNNs for clinical diagnosis.
- **Optimized** model performance via Bayesian hyperparameter tuning, achieving 93% accuracy and reducing false negatives by 22%.
- **Deployed** a real-time inference API using FastAPI and Docker on AWS EC2, achieving sub-200ms latency for rapid reporting.
- **Tech Stack**: Python, PyTorch, FastAPI, Torchvision, Docker, Git, GitHub, AWS.

Crop Yield Prediction System

July 2025 – September 2025

- **Engineered** a predictive analytics pipeline integrating 10 years of historical data across 10+ crops, achieving 96% forecasting accuracy.
- **Automated** complex ETL workflows for 10,000+ records using Pandas and SQL, reducing data preprocessing time by 35%.
- **Developed** a CatBoost regression model, outperforming baseline algorithms by 18% in Mean Absolute Error (MAE) for yield estimation.
- **Tech Stack**: Python, SQL, Machine Learning, CatBoost, Streamlit, Pandas, NumPy, GitHub.

Achievements & Certifications

- **Secured** global rank 1,296 (Top 1.6% of 82,787 teams) in the **Amazon ML Challenge**, engineering scalable predictive models for large-scale datasets. **September 2025 – October 2025**
- **Gained** hands-on expertise in NLP and data modeling through direct mentorship from scientists at the exclusive **Amazon ML Summer School**. **June 2025 – July 2025**
- **Obtained** the **Oracle Cloud Infrastructure Foundations Associate** certification, demonstrating proficiency in core cloud concepts. **October 2025 – November 2025**

Positions of Responsibility

Event Management Lead
BITSA, NIT Raipur — August 2024 – May 2025

- **Strategic Planning & Execution:** Led end-to-end planning and execution of **8+ major departmental events**, implementing targeted engagement strategies that drove a **25% increase** in student participation and improved overall event outreach metrics.
- **Cross-Functional Team Leadership:** Managed and mentored a diverse team of **15+ volunteers** across logistics, marketing, and design functions, streamlining operational workflows and ensuring **100% on-time execution** with zero critical failures.
- **Operations & Data Management:** Optimized event registration workflows and attendee data tracking systems, successfully overseeing end-to-end logistics and database management for **1,200+ total participants**, ensuring seamless data collection and post-event reporting.